Data structure assignment -1

1. Sum of Two Matrices

```
#include<stdio.h>
int main()
{
    int a[10][10], b[10][10], c[10][10];
    int i,j,r,c1;
    printf("Enter rows and columns: ");
    scanf("%d%d",&r,&c1);
    printf("Enter first matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c1;j++)
            scanf("%d",&a[i][j]);
    printf("Enter second matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c1;j++)
            scanf("%d",&b[i][j]);
    printf("Sum Matrix:\n");
    for(i=0;i<r;i++)
{
        for(j=0;j<c1;j++)
        {
            c[i][j]=a[i][j]+b[i][j];
            printf("%d ",c[i][j]);
        }
        printf("\n");
    }
```

```c
    return 0;
}
```

Output:

```
Enter rows and columns: 2 2
Enter first matrix:
1 2
3 4
Enter second matrix:
5 6
7 8
Sum Matrix:
6 8
10 12
```

2. Display Array in Reverse Order

```c
#include<stdio.h>
int main()
{
    int a[100],n,i;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    printf("Reverse order:\n");
    for(i=n-1;i>=0;i--)
        printf("%d ",a[i]);
   return 0;
}
```

Output:

```
Enter number of elements: 3
Enter elements:
50 60 70
Reverse order:
70 60 50
```

3. Count Total Duplicate Elements

```
#include<stdio.h>
int main()
{
    int a[100],n,i,j,count=0;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    for(i=0;i<n;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(a[i]==a[j])
            {
                count++;
                break;
            }
        }
    }
    printf("Total duplicate elements = %d",count);
```

```
    return 0;
}
```

Output:

```
Enter number of elements: 5
Enter elements:
30 40 50 60 50
Total duplicate elements = 1
```

---

4. Print Unique Elements in an Array

```
#include<stdio.h>
int main()
{
    int a[100],n,i,j,flag;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Unique elements are:\n");
    for(i=0;i<n;i++)
    {
        flag=0;
        for(j=0;j<n;j++)
        {
            if(i!=j && a[i]==a[j])
            {
                flag=1;
                break;
```

```
            }
        }
        if(flag==0)
            printf("%d ",a[i]);
    }
    return 0;
}
```

Output:

Enter number of elements: 7

Enter elements:

10 20 30 20 10 40 15

Unique elements are:

30 40 15

---

5. Insert in Sorted Array

```
#include<stdio.h>
int main()
{
    int a[100],n,i,pos,value;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter sorted elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    printf("Enter value to insert: ");
    scanf("%d",&value);
    for(i=0;i<n;i++)
    {
        if(value<a[i])
```

```
        {
            pos=i;
            break;
        }
    }
    for(i=n;i>pos;i--)
        a[i]=a[i-1];

    a[pos]=value;
    printf("Array after insertion:\n");
    for(i=0;i<=n;i++)
        printf("%d ",a[i]);
    return 0;
}
```

Output:

Enter number of elements: 5

Enter sorted elements:

20 25 30 35 40

Enter value to insert: 26

Array after insertion:

20 25 26 30 35 40

---

6. Insert in Unsorted Array

```
#include<stdio.h>
int main()
{
    int a[100], n, i, pos, value;
    printf("Enter number of elements: ");
    scanf("%d",&n);
```

```
    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter position: ");
    scanf("%d",&pos);
    printf("Enter value: ");
    scanf("%d",&value);
    for(i=n;i>=pos;i--)
        a[i]=a[i-1];
    a[pos-1]=value;
    printf("Array after insertion:\n");
    for(i=0;i<=n;i++)
        printf("%d ",a[i]);
    return 0;
}
```

Output:

Enter number of elements: 5

Enter elements:

20 25 30 35 40

Enter position: 3

Enter value: 26

Array after insertion:

20 25 26 30 35 40

---

7. Matrix Multiplication

```
#include<stdio.h>
int main()
{
```

```
int a[10][10], b[10][10], c[10][10];
int i,j,k,r1,c1,r2,c2;
printf("Enter rows and columns of first matrix: ");
scanf("%d%d",&r1,&c1);
printf("Enter rows and columns of second matrix: ");
scanf("%d%d",&r2,&c2);
if(c1!=r2)
{
    printf("Matrix multiplication not possible");
    return 0;
}
printf("Enter first matrix:\n");
for(i=0;i<r1;i++)
    for(j=0;j<c1;j++)
        scanf("%d",&a[i][j]);
printf("Enter second matrix:\n");
for(i=0;i<r2;i++)
    for(j=0;j<c2;j++)
        scanf("%d",&b[i][j]);
for(i=0;i<r1;i++)
{
    for(j=0;j<c2;j++)
    {
        c[i][j]=0;
        for(k=0;k<c1;k++)
            c[i][j]+=a[i][k]*b[k][j];
    }
}
printf("Product Matrix:\n");
```

```
    for(i=0;i<r1;i++)
    {
        for(j=0;j<c2;j++)
            printf("%d ",c[i][j]);
        printf("\n");
    }
    return 0;
}
```

Output:

Enter rows and columns of first matrix: 3 3

Enter rows and columns of second matrix: 3 3

Enter first matrix:

1 2 3

4 5 6

7 8 9

Enter second matrix:

3 5 6

7 9 1

8 0 5

Product Matrix:

41 23 23

95 65 59

149 107 95

---

8. Matrix Transpose

```
#include<stdio.h>
int main()
{
    int a[10][10], i, j, r, c;
```

```c
    printf("Enter rows and columns: ");
    scanf("%d%d",&r,&c);
    printf("Enter matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);
    printf("Transpose Matrix:\n")
    for(i=0;i<c;i++)
    {
        for(j=0;j<r;j++)
            printf("%d ",a[j][i]);
        printf("\n");
    }
    return 0;
}
```

Output:

Enter rows and columns: 3 3

Enter matrix:

1 2 3

4 5 6

7 8 9

Transpose Matrix:

1 4 7

2 5 8

3 6 9

---

9. Right Diagonal Sum

```c
#include<stdio.h>
int main()
```

```
{
    int a[10][10], n, i, j, sum=0;

    printf("Enter order of matrix: ");
    scanf("%d",&n);

    printf("Enter matrix:\n")
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);
    for(i=0;i<n;i++)
        sum+=a[i][n-1-i]
    printf("Right diagonal sum = %d",sum);
    return 0;
}
```

Output:

```
Enter order of matrix: 3
Enter matrix:
1 2 3
4 5 6
7 8 9
Right diagonal sum = 15
```

---

10. Check for Identity Matrix

```
#include<stdio.h>
int main()
{
    int a[10][10], n, i, j, flag=1;
```

```
    printf("Enter order of matrix: ");
    scanf("%d",&n);
    printf("Enter matrix:\n");
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);
    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
        {
            if(i==j && a[i][j]!=1)
                flag=0;

            if(i!=j && a[i][j]!=0)
                flag=0;
        }
    }
    if(flag)
        printf("Identity Matrix");
    else
        printf("Not an Identity Matrix");
    return 0;
}
```

Output:

```
Enter order of matrix: 2
Enter matrix:
1 0
0 1
Identity Matrix
```

11. Count Vowels, Consonants, Digits and Spaces

```
#include<stdio.h>
#include<string.h>
int main()
{
    char str[100];
    int i,v=0,c=0,d=0,s=0;
    printf("Enter a string: ");
    fgets(str,100,stdin);
    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]=='a'||str[i]=='e'||str[i]=='i'||str[i]=='o'||str[i]=='u'||
           str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U')
            v++
        else if((str[i]>='A'&&str[i]<='Z')||(str[i]>='a'&&str[i]<='z'))
            c++;
        else if(str[i]>='0'&&str[i]<='9')
            d++;


        else if(str[i]==' ')
            s++;
    }
    printf("Vowels=%d\nConsonants=%d\nDigits=%d\nSpaces=%d",v,c,d,s);
    return 0;
}
```

Output:

Enter a string: Hello World

Vowels=3

Consonants=7

Digits=0

Spaces=1

---

12. Frequency of a Character in a String

```
#include<stdio.h>
int main()
{
    char str[100], ch;
    int i,count=0;
    printf("Enter string: ");
    fgets(str,100,stdin);
    printf("Enter character: ");
    scanf("%c",&ch);

    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]==ch)
            count++;
    }
    printf("Frequency = %d",count);
    return 0;
}
```

Output:

Enter a string: Hello World

Enter character: l

Frequency = 3

---

13. Count Words in a String

```
#include<stdio.h>
int main()
{
    char str[100];
    int i,words=1;
    printf("Enter sentence: ");
    fgets(str,100,stdin);
    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]==' ')
            words++;
    }
    printf("Number of words = %d",words);
    return 0;
}
```

Output:

Enter sentence: Data Structure

Number of words = 2

---

14. Sort String Array

```
#include<stdio.h>
#include<string.h>
int main()
{
    char str[5][20], temp[20];
    int i,j;
    printf("Enter 5 strings:\n");
    for(i=0;i<5;i++)
        scanf("%s",str[i]);
```

```
    for(i=0;i<4;i++)
    {
        for(j=i+1;j<5;j++)
        {
            if(strcmp(str[i],str[j])>0)
            {
                strcpy(temp,str[i]);
                strcpy(str[i],str[j]);
                strcpy(str[j],temp);
            }
        }
    }
    printf("Sorted strings:\n");
    for(i=0;i<5;i++)
        printf("%s\n",str[i]);
    return 0;
}
```

Output:

```
Enter 5 strings:
Spiderman
Iron man
Captain
Thor
Hulk
Sorted strings:
Captain
Hulk
Iron man
Spiderman
```

Thor

---

15. Toggle Case of a Sentence

```
#include<stdio.h>
int main()
{
    char str[100];
    int i;
    printf("Enter a sentence: ");
    fgets(str,100,stdin);
    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]>='A'&&str[i]<='Z')
            str[i]=str[i]+32;
        else if(str[i]>='a'&&str[i]<='z')
            str[i]=str[i]-32;
    }
    printf("After Toggle Case:\n%s",str);
    return 0;
}
```

Output:

Enter a sentence: My name is Rajesh

After Toggle Case:

mY NAME IS rAJEAH